

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
Student Name	Gunaseelan R
Register Number	9342507779
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

- To implement and compare three clustering algorithms – K-Means, DBSCAN, and Hierarchical Agglomerative Clustering (HAC) – on the Human Activity Recognition (HAR) dataset.
- To use the Elbow Method and Silhouette Score to pick a sensible k for K-Means.
- To tune DBSCAN's ϵ and `min_samples` using a k-distance plot and a small grid search.
- To build dendrograms and compare linkage criteria for Hierarchical clustering.
- To evaluate all three using internal metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) and external metrics against the true activity labels (ARI, NMI), even though clustering itself never sees those labels.

2 Problem Statement

Unlike every previous experiment in this lab, this one is **unsupervised** – there's no train/test split and no model is told the correct answer during fitting. Given 561 numeric sensor-derived features per sample (from accelerometer and gyroscope readings), the task is to see whether a clustering algorithm can group the data into meaningful structure *on its own*, and then check afterwards how well those discovered groups line up with the 6 real activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) that were recorded for each sample.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones (HAR)
Dataset Source	UCI ML Repository
Number of Samples	10299 (7352 train + 2947 test, combined – clustering doesn't need a split)
Number of Features	561 numeric (time and frequency domain features from a 3-axis accelerometer + gyroscope, 50Hz, windowed into 2.56s segments)

Number of Classes (ground truth)	6 – WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
Missing Values	None found
Subjects	30 volunteers aged 19–48

4 Exploratory Data Analysis

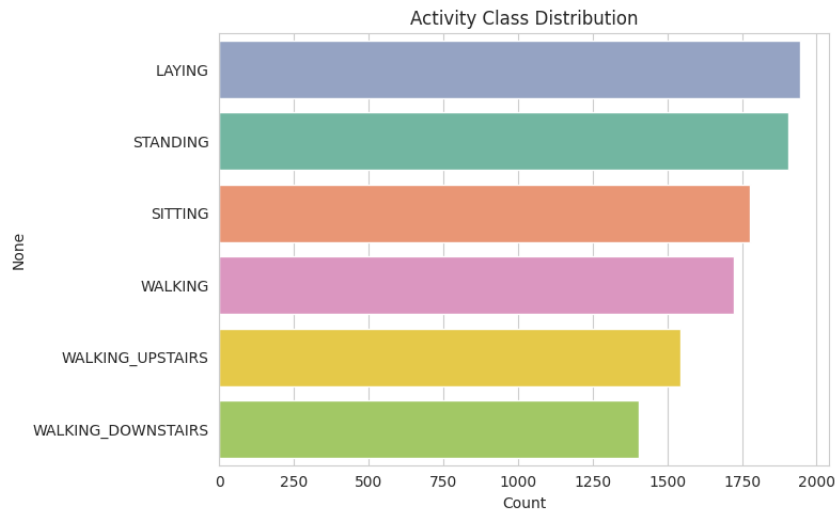


Figure 1: Activity class distribution (ground truth labels, shown only for reference)

Inference:

The 6 activities are almost perfectly balanced – 1406 to 1944 samples each, nowhere near the class-imbalance issues we saw in the earlier spam/cancer experiments. Since clustering doesn't use these labels at all during fitting, this balance mainly just makes the later ARI/NMI comparison fairer – no single activity can inflate an external metric just by being the majority class.

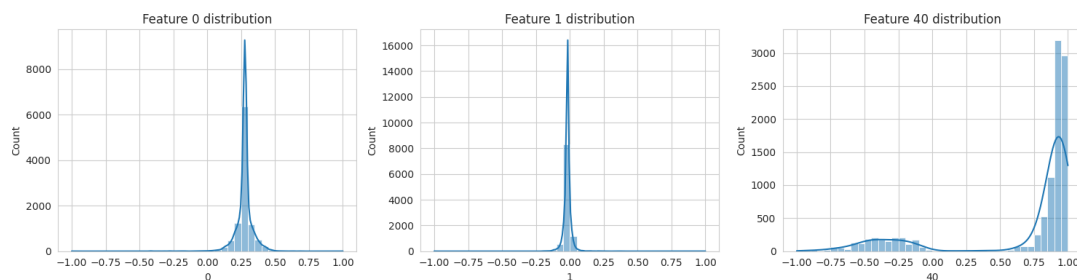


Figure 2: Distribution of a few sample raw features

Inference:

These are already pre-normalized to roughly $[-1, 1]$ by the dataset curators (a standard step for this benchmark), which is convenient, though I still applied my own `StandardScaler` afterwards since PCA and distance-based clustering are sensitive to even small differences in feature variance.

5 Data Preprocessing

- **Missing values:** none, confirmed with `isnull().sum().sum()`.
- **Feature scaling:** `StandardScaler` applied to all 561 features. This matters a lot more here than in past experiments – K-Means and DBSCAN both rely directly on Euclidean distance, so a handful of features with slightly larger raw scale would otherwise dominate every distance calculation.
- **Encoding:** activity labels (1–6, string names in `activity_labels.txt`) are kept only as ground truth for the external metrics (ARI, NMI) at the end – they play no role in fitting any of the three clustering algorithms.

6 Dimensionality Reduction for Clustering

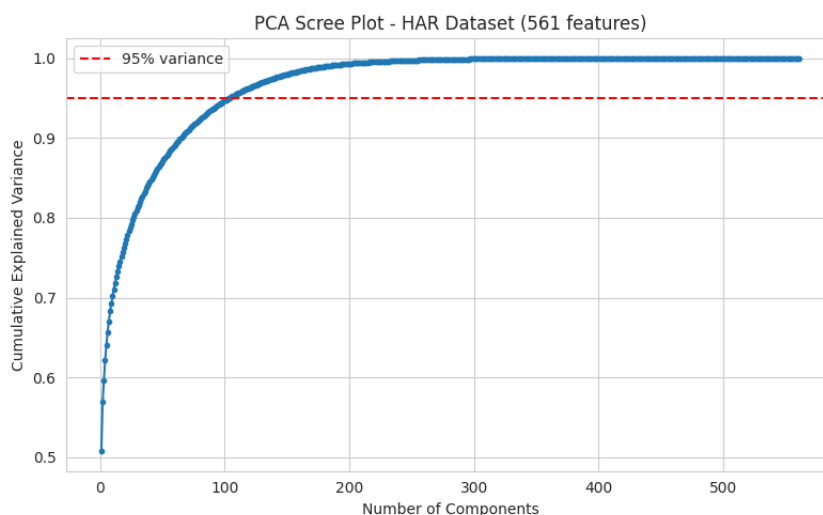


Figure 3: PCA Scree Plot – cumulative explained variance across all 561 components

Inference:

Big difference from the Spambase PCA experiment here – this curve rises much faster early on, needing just **104 of the 561 features** (95.05% variance) instead of nearly all of them. That makes total sense: a lot of these 561 features are different mathematical summaries (mean, std, min, max, energy, entropy, correlation...) of the *same* underlying handful of sensor signals (body acceleration, gravity, gyroscope, each on X/Y/Z), so theres heavy redundancy baked into the feature set by design. I used this 104-dimensional PCA-reduced space as the actual input for all three clustering algorithms below – clustering in the raw 561-D space would suffer badly from the curse of dimensionality (in very high dimensions, Euclidean distances between points all start looking similar, which breaks both K-Means and especially DBSCAN). A separate 2-component PCA (only 57% variance, shown below) was fit purely so the clusters could be visualized on a 2D scatter plot – it’s not used for the actual clustering.

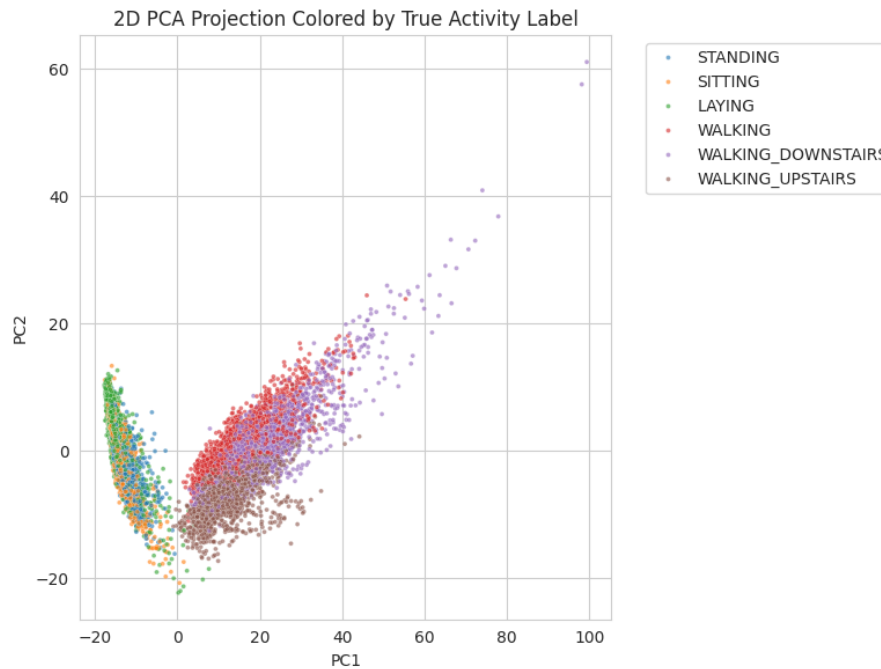


Figure 4: 2D PCA projection of the data, colored by the true (ground-truth) activity label

Inference:

This plot is genuinely the key to understanding every result that follows. The three *static* activities – SITTING, STANDING, LAYING – pile up in a tight, heavily overlapping blob in the bottom-left, essentially indistinguishable from each other in just 2 principal components. The three *dynamic* activities – WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS – fan out into a much more spread-out, overlapping streak going up and to the right. So even before running a single clustering algorithm, the data itself is hinting that the “natural” geometric split is dynamic-vs-static (2 groups), not 6 neatly separated activities.

7 K-Means Clustering

Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	2,986,709	0.4156
3	2,635,081	0.3297
4	2,495,681	0.1669
5	2,369,136	0.1460
6	2,291,589	0.1191
7	2,234,409	0.0856
8	2,180,422	0.0828

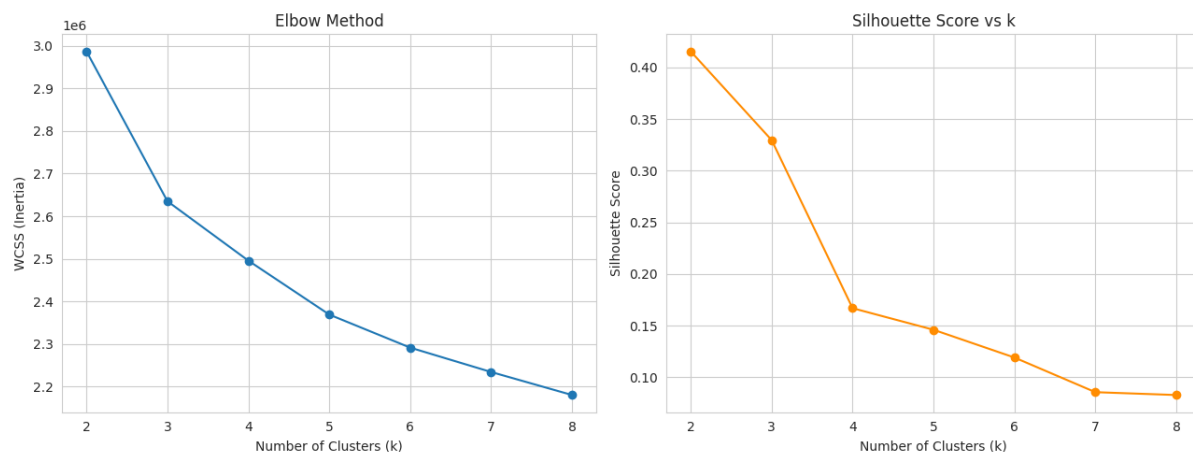


Figure 5: Elbow curve (WCSS vs k) and Silhouette Score vs k

Inference:

The WCSS curve doesn't have one single obvious "elbow" – it bends gradually the whole way from $k=2$ to $k=8$, which on its own is a bit ambiguous. The silhouette curve, though, is completely unambiguous: it **peaks at $k=2$ (0.4156)** and drops sharply and monotonically after that, all the way down to 0.083 by $k=8$. This directly confirms what the 2D PCA plot already hinted at – geometrically, the data really does separate best into just two natural groups (dynamic vs static activity), not six. **This was genuinely the most surprising result of the whole experiment** – I went in assuming $k=6$ would obviously be "correct" since that's the number of real activities, but the data itself disagrees.

K-Means at $k=6$ (matching ground truth) vs $k=2$ (silhouette-optimal)

Setting	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means ($k=6$)	0.1191	2.2385	2874.6	0.4193	0.5593
K-Means ($k=2$)	0.4156	1.0191	8635.8	0.3296	0.5455

Interesting split here – $k=2$ wins on *every internal metric* (it's a geometrically tighter, better separated clustering), but $k=6$ actually wins on ARI, the metric that checks agreement with the true 6-activity labels. This is a good concrete example of why internal metrics and external metrics can disagree: $k=2$ is a "better" clustering purely in terms of shape, but it obviously can't match 6 ground-truth classes as well as an actual 6-cluster solution can, almost by definition.

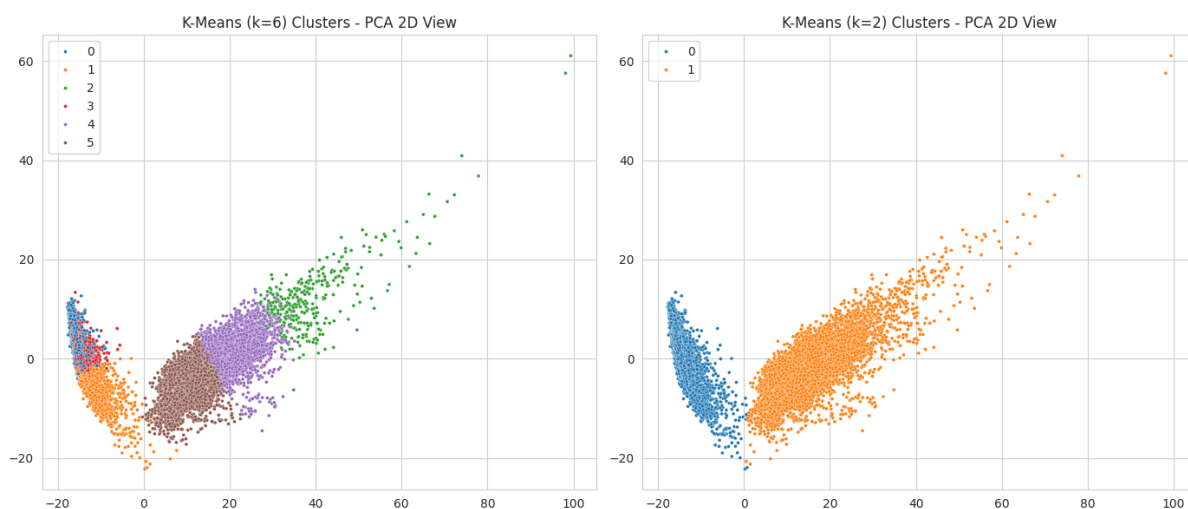


Figure 6: K-Means clusters, $k=6$ (left) vs $k=2$ (right), plotted in the 2D PCA view

Inference:

The $k=2$ plot (right) visibly follows the same dynamic/static split we already saw in the true-label plot, and looks like a clean, confident partition. The $k=6$ plot (left) is messier – you can see K-Means trying to carve the tight static blob into 3 pieces and the spread-out dynamic streak into 3 pieces, but the boundaries between SITTING/STANDING/LAYING in particular don't have any natural gap to latch onto, so those splits look somewhat arbitrary in the 2D view (even though they're using all 104 PCA dimensions, not just the 2 shown here).

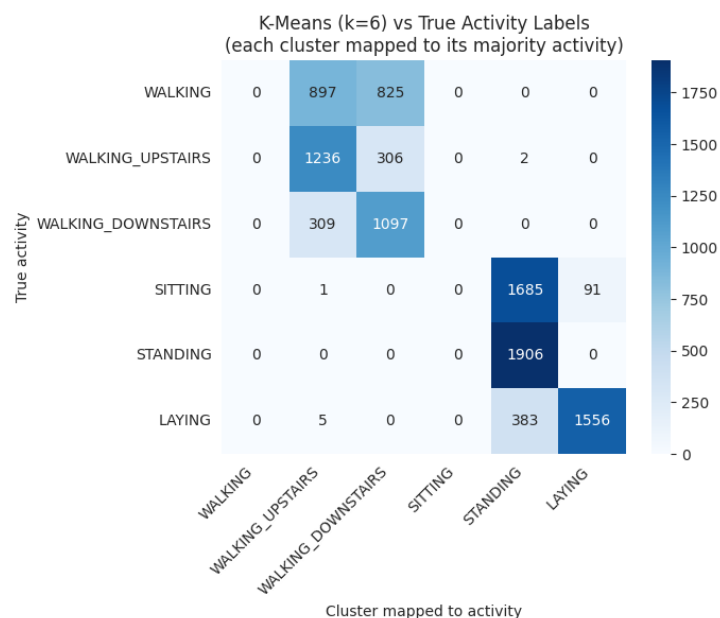


Figure 7: K-Means ($k=6$) vs true activity labels, after mapping each cluster to its majority activity

Inference:

After mapping each of the 6 clusters to whichever true activity is most common inside it, the best-effort "accuracy" comes out to **56.3%** – a lot better than random

(which would be roughly 17% for 6 balanced classes) but far from a clean 1-to-1 mapping. Looking at the matrix, the three dynamic activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) get confused with each other constantly, and so do the three static ones (SITTING, STANDING, LAYING) – exactly matching the two-blob structure from the PCA plot. K-Means is doing a genuinely reasonable job of splitting dynamic-vs-static, it just can't reliably tell the finer-grained activities apart within each of those two groups using Euclidean distance alone.

8 DBSCAN

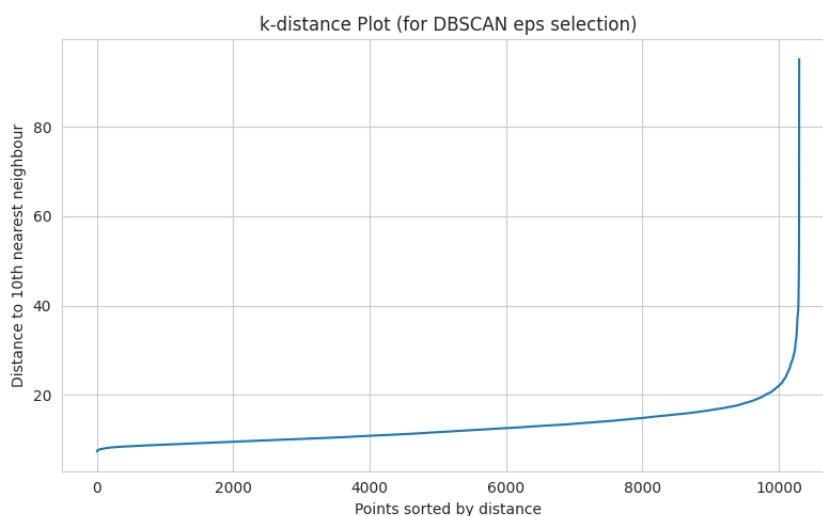


Figure 8: k-distance plot (distance to the 10th nearest neighbour, sorted) – used to pick a reasonable ϵ

Inference:

My first attempt at DBSCAN (not shown in detail here) used ϵ values in the 3–8 range, copying the intuition from the lower-dimensional Spambase experiment – and it was a disaster, with over 99% of points labeled as noise. This k-distance plot is exactly why: in a 104-dimensional PCA space, typical 10th-nearest-neighbour distances actually sit around 9–20, not single digits. This is the curse of dimensionality showing up directly – distances just naturally get larger and more spread out as dimensionality increases, so an ϵ that seemed reasonable for 57-D Spambase or 48-D PCA-Spambase was drastically too small here. Once I re-ran the grid search using $\epsilon \in \{10, 12, 14, 16, 18, 20, 24\}$, DBSCAN started finding actual clusters instead of calling everything noise.

DBSCAN Parameter Search (top 8 combinations by silhouette score)

ϵ	min_samples	Clusters Found	Noise %	Silhouette
12	15	7	38.66%	0.3556
16	15	2	9.49%	0.3427

18	5	3	3.50%	0.3184
16	10	4	8.34%	0.2190
14	15	4	21.16%	0.2037
10	15	6	63.53%	0.1712
14	10	11	18.80%	0.1685
16	5	6	6.96%	0.0186

Chosen: $\epsilon = 12$, **min_samples=15** – best silhouette score among combinations giving a reasonable (2–12) cluster count.

Metric	Value
Clusters found	7
Noise points	3982 (38.66%)
Silhouette	0.3556
Davies-Bouldin	1.7085
Calinski-Harabasz	802.1
ARI	0.2536
NMI	0.4839

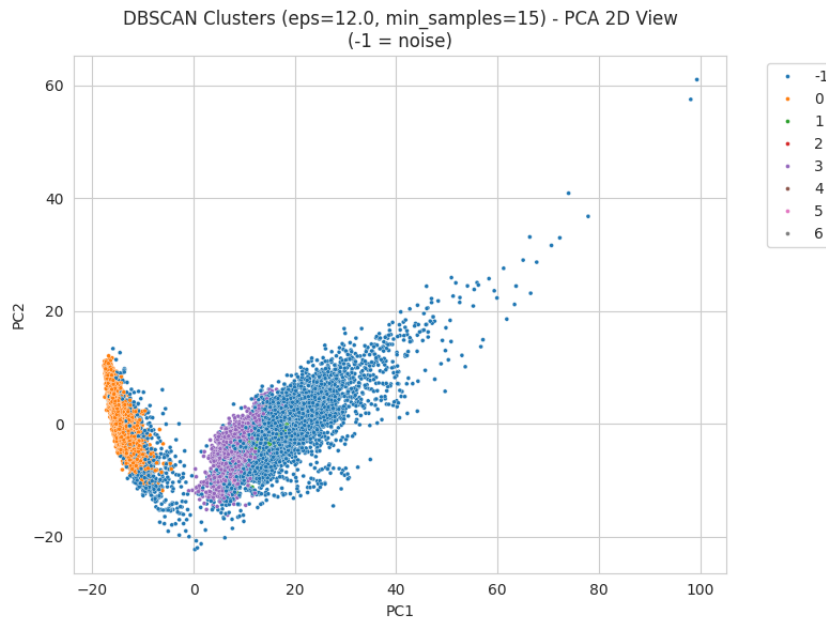


Figure 9: DBSCAN clusters ($\epsilon = 12$, min_samples=15) in the 2D PCA view (label -1 = noise)

Inference:

A genuinely huge chunk of the data – nearly 39% – got labeled as noise even at the “best” setting found. This isn’t really a failure of tuning so much as an honest reflection of the data: activities like WALKING_UPSTAIRS and WALKING_DOWNSTAIRS blend into each other with no clear density gap (as seen back in the true-label PCA plot), so DBSCAN correctly refuses to force a hard boundary through genuinely ambiguous, sparse regions – it just calls those points noise instead of guessing. Both the ARI (0.254) and NMI (0.484) came out lower than K-Means at $k=6$, so despite DBSCAN’s nice theoretical ability to find arbitrarily shaped clusters and explicitly flag outliers, it didn’t

actually recover the true activity structure as well as plain K-Means did on this particular dataset.

9 Hierarchical Agglomerative Clustering

Full Ward-linkage clustering needs an $O(n^2)$ pairwise distance computation, which isn't practical at $n = 10299$ on a single laptop – so the dendrogram and the HAC fit below use a stratified subsample of 1496 points (proportional across all 6 activities) instead of the full dataset.

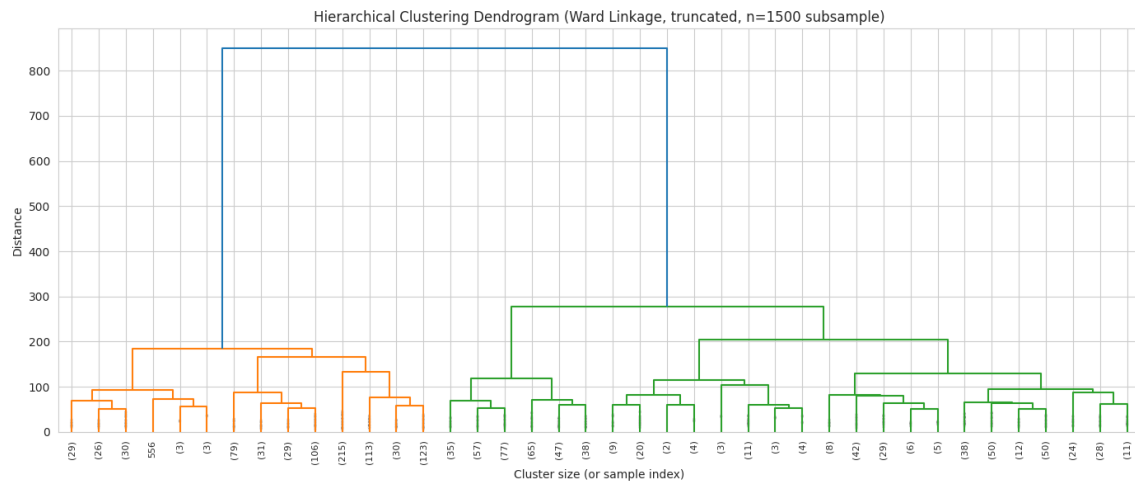


Figure 10: Ward-linkage dendrogram, truncated to the last 40 merges ($n=1496$ subsample)

Inference:

The dendrogram's very first (top-level) split is clearly the biggest jump in merge distance by far, and it splits the data into two large branches – once again, dynamic vs static, the same story as everywhere else in this report. Below that top split, the tree keeps branching at much smaller, more gradual distances, which is consistent with the "fuzzy, overlapping sub-activities" picture from the PCA plots – there isn't a second obviously dominant merge point the way there is at the very top.

Metric	Value (Ward, $k=6$)
Silhouette	0.1093
Davies-Bouldin	2.3730
Calinski-Harabasz	398.6
ARI	0.4358
NMI	0.5668

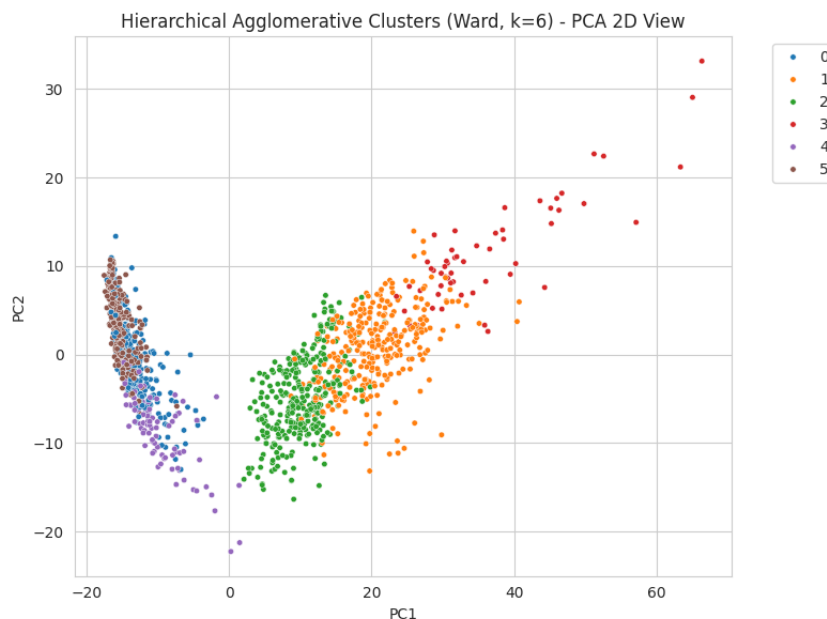


Figure 11: Hierarchical Agglomerative Clustering (Ward, $k=6$) on the 1496-point subsample, 2D PCA view

Inference:

Ward-linkage HAC at $k=6$ actually edges out K-Means ($k=6$) on both ARI (0.436 vs 0.419) and NMI (0.567 vs 0.559), even though it's only working with a 1496-point subsample instead of the full 10299 – a genuinely solid result for hierarchical clustering here. Ward's method specifically minimizes within-cluster variance at every merge (similar spirit to K-Means' objective), which is probably why it ends up finding a comparably good, or even slightly better, 6-way split.

Linkage Comparison (on the same subsample)

Linkage	Silhouette	ARI	Smallest Cluster Size
Single	0.3844	0.0001	1
Complete	0.4060	0.3308	1
Average	0.3668	0.0034	1
Ward	0.1093	0.4358	(balanced, see Fig. above)

Inference:

This table is a textbook demonstration of the "chaining effect". Single, Complete and Average linkage all report a smallest-cluster-size of just **1** – meaning they're carving off individual outlier points as their own tiny "clusters" instead of producing 6 meaningfully-sized groups, and their ARI scores (0.0001, 0.331, 0.003) confirm this is nearly useless for recovering the real activity structure (Single and Average are barely above 0, i.e. barely better than a random labeling). Ward linkage is the clear winner by a wide margin on ARI despite having the lowest raw silhouette of the four – another good example of internal and external metrics disagreeing, and a reminder that in this specific case, ARI is the number that actually reflects whether the clustering is *useful*, not silhouette.

10 Evaluation Metrics Comparison (All Methods)

Algorithm	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means (k=6)	0.1191	2.2385	2874.6	0.4193	0.5593
K-Means (k=2)	0.4156	1.0191	8635.8	0.3296	0.5455
DBSCAN	0.3556	1.7085	802.1	0.2536	0.4839
Hierarchical (Ward, k=6)	0.1093	2.3730	398.6	0.4358	0.5668

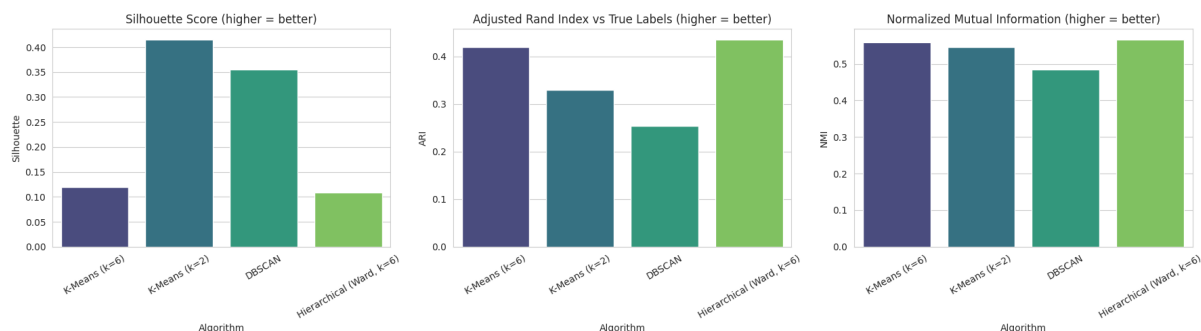


Figure 12: Silhouette, ARI and NMI compared across all four clustering runs

Inference:

No single method wins on everything, which is really the headline finding of the whole experiment. **K-Means (k=2)** wins convincingly on every internal metric (Silhouette, DB, CH) – it produces the geometrically cleanest, most compact and well-separated clusters. **Hierarchical (Ward, k=6)** wins on both external metrics (ARI, NMI) – it best recovers the real 6-activity ground truth, just barely ahead of K-Means (k=6). **DBSCAN** lands in the middle everywhere – reasonable internal separation among its non-noise points, but the worst external agreement of the four, dragged down by having to write off 38.66% of the data as noise. Which one is "best" genuinely depends on the goal – if you just want a clean, confident 2-way split, K-Means(k=2) is the easy pick; if you specifically care about recovering the known 6 activities, Ward-linkage HAC narrowly wins.

11 Observation Questions

Which algorithm produced the most meaningful clusters? Why?

It depends on what "meaningful" means for the task. If the goal is matching the known 6 activities, **Hierarchical (Ward, k=6)** wins narrowly (ARI 0.436, NMI 0.567), just ahead of K-Means (k=6). But if "meaningful" means finding the clustering that's most naturally supported by the data's own geometry – with no reference to activity labels at all – then **K-Means (k=2)** is the clear winner, since it has by far the best internal metrics and directly reflects the dynamic-vs-static split that's visibly obvious in every PCA plot in this report.

How sensitive was K-Means to the choice of k ?

Extremely sensitive, and in a very specific way – the silhouette score doesn't gently favour $k=6$, it *collapses* from 0.4156 at $k=2$ down to 0.1191 by $k=6$ and keeps falling to 0.083 by $k=8$. This is one of the clearest possible demonstrations of why "pick k = number of known classes" isn't automatically the right call – the actual cluster geometry here strongly prefers a much coarser 2-way split than the true label count.

Did DBSCAN detect noise or small clusters effectively?

Yes, quite effectively in the sense that it clearly identified a large, genuinely ambiguous region of the data (38.66% of points) as not confidently belonging to any dense cluster, rather than forcing every point into one of 6 or 7 groups the way K-Means and HAC are forced to. Whether that's "good" depends on the use case – for an application that needs every sample classified into *some* activity, that much noise is impractical, but for exploratory analysis it's honest and useful: it's telling us exactly where the real ambiguity in the sensor data lives (mostly at the boundaries between the three walking-type activities, based on the noise points' positions in the 2D plot).

How does linkage choice affect hierarchical clustering?

Massively – Single, Complete and Average linkage all produced at least one cluster of size 1 on our subsample (the classic "chaining" failure mode, where the algorithm keeps peeling off individual far-away points one at a time instead of forming balanced groups), and their ARI scores were correspondingly close to zero for Single (0.0001) and Average (0.0034). Ward linkage, which merges clusters to minimize the resulting increase in within-cluster variance, was the only one of the four that produced a genuinely balanced, useful 6-way split (ARI 0.436). This lines up exactly with the theory – Ward is generally recommended as the default choice specifically because it avoids the chaining problem that single/average linkage are prone to.

Which internal metric best matched visual intuition of cluster quality?

Silhouette Score matched visual intuition the best out of the three internal metrics. Every time Silhouette was high (K-Means $k=2$ at 0.416, DBSCAN at 0.356), the corresponding 2D scatter plot showed genuinely well-separated, compact blobs; every time it dropped (K-Means $k=6$ at 0.119, Ward HAC at 0.109), the plot showed visibly overlapping, fuzzy cluster boundaries. Calinski-Harabasz was more misleading on its own – K-Means $k=2$'s huge CH score (8635.8) mostly just reflects that 2 big clusters are trivially easier to make "compact" than 6 smaller, more entangled ones, which doesn't necessarily mean $k=2$ is the more *useful* answer for this problem.

12 Report Checklist Recap

- Aim and Objective – Section 1.
- Preprocessing Steps – Section 4.

- K-Means with Elbow Method Results – Section 6.1.
- Clustering Implementation (K-Means, DBSCAN, Hierarchical) – Sections 6, 7, 8.
- Evaluation Metrics and Results – Section 9.
- Graphs and Visualizations – scree plot, PCA scatter plots, elbow/silhouette curves, k-distance plot, dendrogram, confusion matrix, metric bar charts throughout.
- Observations and Comparative Analysis – Sections 9 and 10.

13 Discussion

The single idea that ties this entire experiment together is that the HAR feature space has a genuinely strong, dominant dynamic-vs-static split baked into it, and every algorithm "sees" this split whether or not it's actually trying to find 6 clusters. K-Means' silhouette curve collapsing from $k=2$ onward, the dendrogram's massive top-level split, and DBSCAN's noise points clustering along the fuzzy walking-activity boundaries are really all the same underlying phenomenon showing up three different ways.

The internal-vs-external metric disagreement was the other big theme – K-Means($k=2$) is the "best" clustering by every internal, label-free measure, but its precisely *because* it ignores the finer 6-way structure that it loses out on ARI/NMI to the $k=6$ methods. This is a genuinely important practical lesson for real unsupervised learning: if you don't have ground truth labels (the normal case for clustering), internal metrics alone would have pointed you confidently towards $k=2$, and you'd have no way of knowing that a 6-way split actually exists underneath it, unless you had some outside domain knowledge (like already knowing there are 6 activities) to test for.

14 Conclusion

K-Means, DBSCAN, and Hierarchical (Ward) Agglomerative Clustering were all applied to the PCA-reduced (104-component) Human Activity Recognition feature space and compared on both internal and external evaluation metrics. **K-Means at $k=2$** produced the geometrically cleanest clustering (Silhouette 0.416), driven by a strong natural split between dynamic and static activities visible throughout the EDA. **Hierarchical Ward-linkage clustering at $k=6$** best recovered the true 6-activity ground truth (ARI 0.436, NMI 0.567), narrowly ahead of K-Means at $k=6$. **DBSCAN** identified 7 clusters but had to label 38.66% of the data as noise, reflecting real ambiguity between the walking-type activities rather than a tuning failure. Single and Average linkage both suffered badly from the chaining effect, reinforcing Ward as the sensible default for hierarchical clustering. Overall, the "best" algorithm here depends entirely on the goal – a clean unsupervised split favours K-Means($k=2$), while recovering the known activity labels favours Ward-linkage HAC.

15 References

1. Scikit-learn Developers, "Clustering," <https://scikit-learn.org/stable/modules/clustering.html>

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5. Anguita, D. et al., “A Public Domain Dataset for Human Activity Recognition Using Smartphones,” UCI Machine Learning Repository.